

Youngjoon Kim

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SSLab, School of Cybersecurity and Privacy, Georgia Tech
Security Researcher at Team Atlanta (AIXCC Winner)

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Education

Korea University <i>Ph.D. in Information Security</i> Advisor: Jiwon Yoon <i>Thesis: AI-Driven Automated Vulnerability Discovery and Repair</i>	Sep. 2018 – Feb. 2025 <i>Seoul, Korea</i>
Korea University <i>B.S. in Cyber Defense</i>	Mar. 2013 – Feb. 2017 <i>Seoul, Korea</i>

Experience

Georgia Tech <i>Postdoctoral Researcher</i> Supervisor: Taesoo Kim	Sep. 2025 – Present <i>Atlanta, USA</i>
Korea University <i>Postdoctoral Researcher</i> Supervisor: Jiwon Yoon	Jun. 2025 – Aug. 2025 <i>Seoul, Korea</i>
R.O.K. Cyber Operation Command <i>Security Engineer (R.O.K. Army Captain)</i> - Performed vulnerability assessments for R.O.K. military IT infrastructure. - Worked as a red team during R.O.K. military cyber operation exercises. - Analyzed North Korea's cyberattack techniques and simulated similar attack scenarios for cybersecurity training. - Conducted cybersecurity management ability assessment for public institutions in South Korea.	Oct. 2022 – May. 2024 <i>Seoul, Korea</i>
Agency For Defense Development <i>Security Researcher (R.O.K. Army Captain)</i>	Jul. 2017 – Sep. 2022 <i>Seoul, Korea</i>
Research on National-level cyberattack defense technologies - Developed a predictive model for next-stage attacks using Bayesian network and MITRE ATT&CK. - Implemented network-level and host-level automatic defense using SDN.	Jan. 2021 – Sep. 2022
Research on techniques for evaluating binary fuzzing results - Ported a taint analysis framework from Linux to Windows x64. - Developed a crash triage technique combining directed fuzzing and taint analysis.	Jan. 2018 – Oct. 2020

Publications

Peer-reviewed Papers

- **SoK: DARPA's AI Cyber Challenge (AIXCC): Competition Design, Architectures, and Lessons Learned.**
Cen Zhang, Younggi Park, Fabian Fleischer, Yu-Fu Fu, Jiho Kim, Dongkwan Kim, **Youngjoon Kim**, Qingxiao Xu, Andrew Chin, Ze Sheng, Hanqing Zhao, Michael Pelican, David J. Musliner, Jeff Huang, Jon Silliman, Mikel Mcdaniel, Jefferson Casavant, Isaac Goldthwaite, Nicholas Vidovich, Matthew Lehman, and Taesoo Kim
USENIX Security, 2026 (To appear)
- **OSS-CRS: Liberating AIXCC Cyber Reasoning Systems for Real-World Open-Source Security.**
Andrew Chin, Dongkwan Kim, Yu-Fu Fu, Fabian Fleischer, **Youngjoon Kim**, HyungSeok Han, Cen Zhang, Brian Junekyu Lee, Hanqing Zhao, and Taesoo Kim
20th USENIX WOOT Conference on Offensive Technologies, 2026 (To appear)
- **Logs In, Patches Out: Automated Vulnerability Repair via Tree-of-Thought LLM Analysis.**
Youngjoon Kim, Sunguk Shin, Hyoungshick Kim*, and Jiwon Yoon*
* *Corresponding authors*
USENIX Security, 2025
- **Enhancing Graph of Thought: Enhancing Prompts with LLM Rationales and Dynamic Temperature Control.**
Sunguk Shin and **Youngjoon Kim***
* *Corresponding author*
International Conference on Learning Representations (ICLR), 2025
- **SCVMON: Data-oriented attack recovery for RVs based on safety-critical variable monitoring.**
Sangbin Park, **Youngjoon Kim**, and Donghoon Lee
International Symposium on Research in Attacks, Intrusions, and Defenses (RAID), 2023
- **BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework.**
Youngjoon Kim, Insup Lee, Hyuk Kwon, Gyeongsik Lee, and Jiwon Yoon
IEEE Access, 2023
- **A new approach to training more interpretable model with additional segmentation.**
Sunguk Shin, **Youngjoon Kim** and Jiwon Yoon
Pattern Recognition Letters, 2021
- **Maxaff: Maximizing code coverage with a gradient-based optimization technique.**
Youngjoon Kim and Jiwon Yoon
Electronics, 2020

Under Review

- **CtxFuzz: Context-Aware Directed Fuzzing via Runtime Fuzzer-Agent Cooperation.**
Co-author
Top-tier Software Engineering conference

- **CRSBench: Revealing the Full Capability of Cyber Reasoning Systems via Realistic End-to-End Evaluation.**
First author
Top-tier Security conference
- **SPACEPROTONET: Revealing Unknown Protocols' Origin in Space Communications.**
First author
Security Journal
- **COIN: A Live, Large-scale Benchmark for Semantics-to-Input Code Reasoning.**
Co-author
Top-tier AI conference

On-going Work

- **Codex-Driven Variant Analysis for End-to-End Zero-Day Discovery and Patching at OSS Scale.**
First author
- **Hybridizing Frontier Agents and Small Models for JavaScript Engine Bug Discovery.**
First author
- **Self-Improving Malware Reverse Engineering Agents via Adversarial Learning.**
First author

Preprints

- **ATLANTIS: AI-driven Threat Localization, Analysis, and Triage Intelligence System.**
Taesoo Kim, et al.
arXiv preprint arXiv:2509.14589, 2025

Other Experiences

OSS-CRS	Sep. 2025 – Present
<i>Main Contributor</i>	
AIxCC Final (1st Place)	Dec. 2024 – Aug. 2025
<i>Member of Team ATLANTA</i>	
AIxCC Semi-final	Apr. 2024 – Aug. 2024
<i>Tech Leader of Team KORIA</i>	